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Gruppe: 1 ~~2~~

Punkte: / 24

I2C - TEMPERATURSENSOR

AUFGABE

Programmieren Sie die Ansteuerung des I2C-Temperatursensors MCP9803 am APUS-Board. Die aktuelle Temperatur (auf 0,5°C genau) soll zyklisch auf dem Display angezeigt bzw. aktualisiert werden. Programmieren Sie zusätzlich TSET und THYST Werte (Active-low, Comparator-Mode) nach Ihren Vorstellungen und steuern Sie mit dem Alert-Output (per Interrupt) eine beliebige LED an.

Erstellen Sie dazu einen eigenen Treiber (.c und .h). Folgende Funktionen soll der Treiber minimal unterstützen:

- Temperatursensor initialisieren (und damit auch I2C) und konfigurieren (z.B. Alert)
- Temperaturwert auslesen
- Schwellwerte TSET und THYST setzen

HINWEISE

- Der I2C-Treiber muss zum Teil noch finalisiert werden.
- Achten Sie darauf, die korrekten PINs für I2C zu initialisieren (SDA und SCL), da es hierbei mehrere Möglichkeiten gibt.
- Der MCP9803 hat ein I2C-Timeout von 35ms, daher kann der Code nicht per Single-Step debugged werden.
- Aktivieren Sie I2C in der Datei stm32f0xx_conf.h und inkludieren Sie stm32f0xx_i2c.c

FRAGEN

- Wofür wird FAULT QUEUE verwendet?
- Wie viele Temperatursensoren können parallel maximal am I2C-Bus betrieben werden?
- Welche Auflösung des Sensors ist nach dem Reset eingestellt?
- Wozu kann der One-Shot Mode verwendet werden?

ABGABE

- Sourcecode (16 Punkte)
- Beantwortete Fragen (4 Punkte)
- Funktionsbeschreibung der Software und ein Foto des Displays im Betrieb (4 Punkte)

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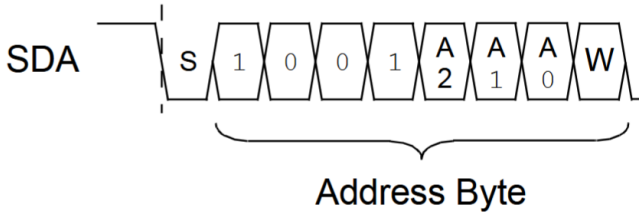
1 Fragen

1.1 Wofür wird FAULT QUEUE verwendet?

Die *Fault Queue* dient als Filter, um Fehlauslösungen des `ALERT`-Pins zu reduzieren. Dazu muss die gemessene Temperatur T_A für eine einstellbare Anzahl aufeinanderfolgender Wandlungszyklen oberhalb von T_{SET} liegen, bevor `ALERT` aktiviert wird. Entsprechend muss T_A auch für die eingestellte Anzahl von Zyklen unterhalb von T_{HYST} liegen, bevor `ALERT` wieder deaktiviert wird bzw. ein neuer Interrupt ausgelöst werden kann.

1.2 Wie viele Temperatursensoren können parallel maximal am I2C-Bus betrieben werden?

Am I^2C-Bus können maximal **acht** MCP9800/1/2/3-Sensoren parallel betrieben werden, da es **3** adressierbare Pins (A0, A1, A2) gibt, die jeweils mit GND oder VDD verbunden werden können. Dadurch ergeben sich $2^3 = 8$ mögliche Adresskombinationen für die Sensoren.



1.3 Welche Auflösung des Sensors ist nach dem Reset eingestellt?

Nach dem Reset bzw. nach Power-up ist standardmäßig eine Auflösung von **9 Bit** eingestellt. Das entspricht einer Temperaturauflösung von **0,5 °C**.

bit 5-6	ΣΔ **ADC RESOLUTION** bits
	00 = 9 bit or 0.5°C (Power-up default)
	01 = 10 bit or 0.25°C
	10 = 11 bit or 0.125°C
	11 = 12 bit or 0.0625°C

1.4 Wozu kann der One-Shot Mode verwendet werden?

Der *One-Shot Mode* wird für **stromsparende Anwendungen** verwendet. In diesem Modus führt der Sensor auf Anforderung genau **eine einzelne Temperaturmessung** durch und kehrt danach automatisch wieder in den *Shutdown Mode* zurück. Das ist besonders nützlich, wenn die Temperatur nur gelegentlich und gezielt durch einen Controller abgefragt werden soll.

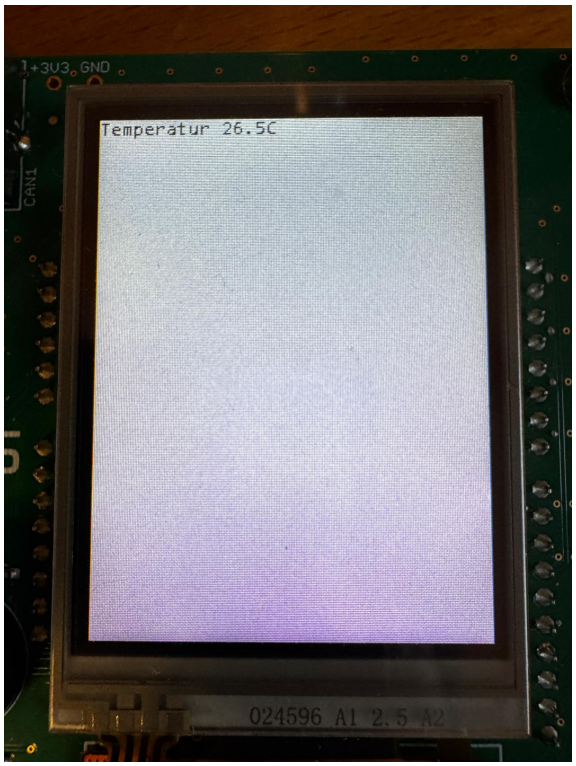
2 Funktionsbeschreibung

Im Projekt wird die aktuelle Temperatur über I2C vom MCP9800 Sensor ausgelesen und auf einem LCD-Display angezeigt. Zusätzlich wird ein Alarm ausgelöst, wenn die gemessene Temperatur den eingestellten Schwellenwert überschreitet. Der `ALERT`-Pin des Sensors ist mit einem externen Interrupt verbunden, der bei Aktivierung eine LED einschaltet, um den Alarm visuell anzuzeigen. Die Temperaturwerte werden kontinuierlich aktualisiert. Bei aktivem Alarm (Temperatur über T_{SET}) leuchtet die LED4 auf, und bei Rückkehr unter T_{HYST} erlischt sie wieder. Die Einstellung von T_{SET} und T_{HYST} erfolgt über das **i2c_temp.h** File.

2.1 MCP Driver

Der Treiber für den MCP9800 Temperatursensor baut auf den I2C Treiber auf. Dieser ist so ausgelegt, dass mehrere Sensoren parallel am I2C-Bus betrieben werden können. Dazu muss für jeden Sensor ein eigener Handle angelegt werden. Im Handle werden im Bereich **init** der Initialisierungszustand des MCP9800 gesetzt und anschließend mit der Funktion `MCP9800_Temp_Init` initialisiert. Die Initialisierung ist mittels Enumerationen der einstellbaren Parameter realisiert, dies ermöglicht eine einfache Typprüfung des Compilers und verhindert ungültige Werte für die Parameter. Bei jedem Aufruf einer Treiber Funktion ist der entsprechende Handle zu übergeben. Status der Funktion wird über den Rückgabewert der Funktion zurückgegeben. Hier wird die I2C_RetType verwendet, um Fehler zu erkennen und entsprechend zu reagieren.

2.2 Bild vom Display



3 Code

3.1 main.c

```c
/**
  ******************************************************************************
  * @file    main.c
  * @author  Simon Offenberger
  * @version V1.0
  * @date    14-October-2025
  * @brief   This file Contains the main Function which calles the current poject main
  ******************************************************************************
  */

/* Includes ------------------------------------------------------------------*/
#include "stm32f0xx_conf.h"
#include "sysdelay.h"
#include "3bit_cnt.h"
#include "alarm.h"
#include "LedTimer.h"
#include "Interrupt_Proj.h"
#include "Uart_Vcom.h"
#include "Uart_DMA.h"
#include "ADC_Temp_prj.h"
#include "i2c_temp.h"

/* Private typedef -----------------------------------------------------------*/
/* Private define ------------------------------------------------------------*/
/* Private macro -------------------------------------------------------------*/
/* Private variables ---------------------------------------------------------*/
/* Private function prototypes -----------------------------------------------*/
/* Private functions ---------------------------------------------------------*/

/**
  * @brief  Initialize the sys tick timer (M0 core) which is used for delay.
  * @param  None
  * @retval None
  */
void SysTick_Init(void)
{
    /* init the sys tick timer to be called every 1ms */
    SysTick_Config(SystemCoreClock / 1000);
}

/**
  * @brief  This is the SysTick interrupt handler which is called every 1ms.
           We have to increment the HAL tick counter which is used for SysDelay
  * @param  None
  * @retval None
  */
void SysTick_Handler()
{
    SysDelay_IncTicks();
}

/**
* @brief  Main program
* @param  None
* @retval None
*/
int main(void)
{
    /* init the sys tick counter (needed for SysDelay) */
    SysTick_Init();

    // switch to project I2C Temperature Sensor
    main_I2C_Temp();
}
```

3.2 i2c_temp.h

```c
/**
 * @file i2c_temp.h
 * @author Simon Offenberger
 * @brief I2C Temperature Sensor Project which indicates a Alert via LED4 and shows the current temperature on the LCD.
 * @version 1.0
 * @date 2026-03-09
 *
 * @copyright Copyright (c) 2026
 *
 */

#ifndef __MAIN_I2C_TEMP_H
#define __MAIN_I2C_TEMP_H

#include <stdint.h>

/** @brief Temperature polling interval in milliseconds. */
#define I2C_POLL_INTERVAL_MS 250

/** @brief Text buffer size for LCD output. */
#define LCD_TEXT_BUFFER_SIZE 32

/** @brief Alert threshold temperature in degree Celsius. */
#define TEMP_ALERT_THRESHOLD  28.5f
/** @brief Alert hysteresis temperature in degree Celsius. */
#define TEMP_ALERT_HYSTERESIS 27.0f

/**
 * @brief Main function for the Project I2C Temperature Sensor.
 */
void main_I2C_Temp(void);

#endif /* __MAIN_I2C_TEMP_H */
```

3.3 i2c_temp.c

```c
/**
 * @file i2c_temp.c
 * @author Simon Offenberger
 * @brief I2C Temperature Sensor Project which indicates a Alert via LED4 and shows the current temperature on the LCD.
 * @version 1.0
 * @date 2026-03-09
 *
 * @copyright Copyright (c) 2026
 *
 */

#include "board_led.h"
#include "sysdelay.h"
#include "i2c_temp.h"
#include "board_MCP_temp.h"
#include "Lcd.h"
#include <stdio.h>

// includes for STM32F0xx Standard Peripheral Library
#include "stm32f0xx.h"
#include "stm32f0xx_gpio.h"
#include "stm32f0xx_exti.h"
#include "stm32f0xx_syscfg.h"
#include "stm32f0xx_rcc.h"

/**
 * @brief  Configures EXTI line 4 for MCP9800 alert interrupt on PD4.
 */
void Alert_Initialize_EXTI(void)
{
  // Enable peripheral clocks
  // Peripherals are not accessible until their clocks are enabled.
  // SYSCFG is needed to map EXTI lines to GPIO pins !
  RCC_APB2PeriphClockCmd(RCC_APB2Periph_SYSCFG, ENABLE);
  RCC_AHBPeriphClockCmd(RCC_AHBENR_GPIODEN, ENABLE);

  GPIO_InitTypeDef initStruct;

  // Load default values into structure
  GPIO_StructInit(&initStruct);

  // Configure alert input pin for external interrupt
  initStruct.GPIO_Pin = GPIO_Pin_4;
  initStruct.GPIO_Mode = GPIO_Mode_IN;
  initStruct.GPIO_PuPd = GPIO_PuPd_UP; // Alert pin is active low, so we use pull-up (external PU also populated)
  GPIO_Init(GPIOD, &initStruct);

  // Connect EXTI lines to GPIO pins
  SYSCFG_EXTILineConfig(EXTI_PortSourceGPIOD, EXTI_PinSource4); // EXTI4 -> PD4

  // Configure EXTI lines
  EXTI_InitTypeDef extiInitStruct;
  extiInitStruct.EXTI_Line = EXTI_Line4;
  extiInitStruct.EXTI_Mode = EXTI_Mode_Interrupt;
  extiInitStruct.EXTI_Trigger = EXTI_Trigger_Rising_Falling; // Interrupt on both edge (alert asserted)
  extiInitStruct.EXTI_LineCmd = ENABLE;
  EXTI_Init(&extiInitStruct);

  // Enable interrupt request in NVIC
  NVIC_EnableIRQ(EXTI4_15_IRQn);
}

/**
 * @brief  Handles EXTI Line4 interrupt generated by MCP9800 alert.
 */
void EXTI4_15_IRQHandler(void)
{
  // Check if EXTI Line4 has a pending interrupt.
  if (EXTI->PR & EXTI_PR_PR4)
  {
    // Clear the EXTI Line4 pending bit no read modify write!!!
    // Because it would reset another pending flag as well!
    // 1 written to the pending register clears the pending bit
    EXTI->PR = EXTI_PR_PR4;

    // Set LED to signal Alert
    if(GPIO_ReadInputDataBit(GPIOD, GPIO_Pin_4) == Bit_RESET){
      LED_On(LED4);
    }
    else{
      LED_Off(LED4);
    }
  }
}

/**
 * @brief Main function for the Project I2C Temperature Sensor Exercise 8.
 */
void main_I2C_Temp()
{
  // Enable global interrupts
  __enable_irq();

  // Initialize Peripheral
  LED_Initialize();
  Lcd_Initialize();
  Lcd_Fill(0, 0, LCD_WIDTH, LCD_HEIGHT, RGB565_WHITE);

  // Initialize MCP9800 temperature sensor with user-selected configuration.
  const MCP9800_Init_StructTypeDef mcp9800Init = {
```

```
        .Shutdown_Mode = MCP9800_SHUTDOWN_MODE_DISABLE,
        .Alert_Mode = MCP9800_COMPARATOR_MODE,
        .Alert_Polarity = MCP9800_ALERT_POLARITY_ACTIVE_LOW,
        .Fault_Queue = MCP9800_FAULT_QUEUE_1,
        .Resolution = MCP9800_ADC_RESOLUTION_0_5C,
        .One_Shot = MCP9800_ONE_SHOT_DISABLE};

    MCP9800_Temp_HandleTypeDef MCP9800_Handle = {
        .i2c_handle = I2C_RTC_TEMP,
        .adr = MCP9800_SLAVE_ADDRESS,
        .init = mcp9800Init};

    MCP9800_Temp_Init(&MCP9800_Handle);

    // Program alert upper threshold and hysteresis values.
    MCP9800_Temp_Set_TSET(&MCP9800_Handle, TEMP_ALERT_THRESHOLD);
    MCP9800_Temp_Set_THYST(&MCP9800_Handle, TEMP_ALERT_HYSTERESIS);

    // Initialize EXTI for Alert pin to signal overtemperature events via LED4.
    Alert_Initialize_EXTI();

    while (1)
    {
      float temp = 0;

      // Periodically poll sensor temperature via I2C.
      MCP9800_Temp_ReadTemp(&MCP9800_Handle, &temp);

      char temp_str[LCD_TEXT_BUFFER_SIZE] = {0};

      // Update current temperature on LCD.
      snprintf(temp_str, LCD_TEXT_BUFFER_SIZE, "Temperatur␣%3.1fC", temp);

      Lcd_Print(0, 0, temp_str, RGB565_BLACK, RGB565_WHITE, &Consolas);

      SysDelay_Delay(I2C_POLL_INTERVAL_MS);
    }
}
```

4 Module

4.1 board_i2c.h

```c
/**
 ******************************************************************************
 * @file    i2c.h
 * @author  Andreas Oyrer
 * @version V1.0
 * @date    10-March-2022
 * @brief   APUS Board I2C control Header File.
 ******************************************************************************
 */

#ifndef __I2C_H
#define __I2C_H

/* Includes ------------------------------------------------------------------*/
#include <stdint.h>
#include "stm32f0xx_i2c.h"

/* Exported types ------------------------------------------------------------*/
#define I2C_RTC_TEMP I2C1

typedef enum
{
  I2C_OK = 0,
  I2C_ERR_NACK,
  I2C_ERR_SLAVE_ADDRESS,
  I2C_ERR_INVALID_PARAM,
  I2C_ERR_TIMEOUT
} I2C_RetType;

/* Exported functions ------------------------------------------------------- */

/**
 * @brief  Initialize I2C including GPIO ports.
 * @param  I2C_Handle Pointer to I2C peripheral instance.
 * @retval The result of the operation.
 */
I2C_RetType I2C_Initialize(I2C_TypeDef *I2C_Handle);

/**
 * @brief  Write data to a i2c slave device.
 * @param  I2C_Handle Pointer to I2C peripheral instance.
 * @param  slave_address Address of the I2C slave device.
 * @param  sendData Data to be sent to the I2C slave device.
 * @param  bytesToSend Number of bytes to send to the I2C slave device.
 * @retval The result of the operation.
 */
I2C_RetType I2C_Write(I2C_TypeDef *I2C_Handle, const uint16_t slave_address, const uint8_t *sendData, const uint8_t
     bytesToSend);

/**
 * @brief  First writes data to a i2c slave device and then reads back the content.
 *         Therefore the first i2c frame is sent without a STOP bit.
 *         The read transfer starts with a repeated START condition.
 * @param  I2C_Handle Pointer to I2C peripheral instance.
 * @param  slave_address Address of the I2C slave device.
 * @param  sendData Data to be sent to the I2C slave device.
 * @param  bytesToSend Number of bytes to send to the I2C slave device.
 * @param  recData Pointer to store received data from the I2C slave device.
 * @param  bytesToRead Number of bytes to read from the I2C slave device.
 * @retval The result of the operation.
 */
I2C_RetType I2C_WriteRead(I2C_TypeDef *I2C_Handle, const uint16_t slave_address, const uint8_t *sendData, const uint8_t
     bytesToSend, uint8_t *recData, const uint8_t bytesToRead);

#endif
```

4.2 board_i2c.c

```c
/**
 ******************************************************************************
 * @file    i2c.c
 * @author  Andreas Oyrer
 * @version V1.0
 * @date    10-March-2022
 * @brief   APUS Board I2C control Source File.
 ******************************************************************************
 */

/* Includes ------------------------------------------------------------------*/
#include "stm32f0xx_gpio.h"
#include "stm32f0xx_rcc.h"
#include "i2c.h"
#include "sysdelay.h"

/* Private typedef -----------------------------------------------------------*/
/* Private define ------------------------------------------------------------*/
#define I2C_TIMEOUT 100

/* Private macro -------------------------------------------------------------*/
/* Private variables ---------------------------------------------------------*/
/* Private function prototypes -----------------------------------------------*/
static int32_t I2C_WaitForFlagOrNackToBeSet(I2C_TypeDef *I2C_Handle,const uint32_t flag);

/**
 * @brief  Initialize I2C including GPIO ports.
 * @param  I2C_Handle Pointer to I2C peripheral instance.
 * @retval The result of the operation.
 */
I2C_RetType I2C_Initialize(I2C_TypeDef *I2C_Handle)
{
    if(I2C_Handle == 0)
    {
        return I2C_ERR_INVALID_PARAM;
    }

    I2C_InitTypeDef initStructI2C;
    GPIO_InitTypeDef initStructGPIO;

    if (I2C_Handle == I2C_RTC_TEMP)
    {
        /* GPIO Ports Clock Enable */
        RCC_AHBPeriphClockCmd(RCC_AHBENR_GPIOBEN, ENABLE);

        GPIO_StructInit(&initStructGPIO);

        /* Configure GPIO pins for I2C: PB6 PB7 */
        initStructGPIO.GPIO_Pin = GPIO_Pin_6 | GPIO_Pin_7;
        initStructGPIO.GPIO_Mode = GPIO_Mode_AF;
        GPIO_Init(GPIOB, &initStructGPIO);
        GPIO_PinAFConfig(GPIOB, GPIO_PinSource6, GPIO_AF_1);
        GPIO_PinAFConfig(GPIOB, GPIO_PinSource7, GPIO_AF_1);

        /* init I2C1 */
        RCC_APB1PeriphClockCmd(RCC_APB1Periph_I2C1, ENABLE);

        I2C_StructInit(&initStructI2C);
        /* Master-Mode, 100kHz, Analog Filter Delay = On, Rise time = 140ns, Fall time = 40ns */
        /* calculated with I2C_Timing_Configuration_V1.0.1.xls */
        initStructI2C.I2C_Timing = 0x10905F86;
        I2C_Init(I2C_Handle, &initStructI2C);
        /* enable I2C */
        I2C_Cmd(I2C_Handle, ENABLE);
    }

    return I2C_OK;
}

/**
 * @brief  Write data to a i2c slave device.
 * @param  I2C_Handle Pointer to I2C peripheral instance.
 * @param  slave_address Address of the I2C slave device.
 * @param  sendData Data to be sent to the I2C slave device.
 * @param  bytesToSend Number of bytes to send to the I2C slave device.
 * @retval The result of the operation.
 */
I2C_RetType I2C_Write(I2C_TypeDef *I2C_Handle, const uint16_t slave_address, const uint8_t *sendData, const uint8_t
    bytesToSend)
{
    uint32_t idx = 0;
    int32_t ret = 0;

    /* null pointer check */
    if (I2C_Handle == 0 || sendData == 0 || bytesToSend == 0)
    {
        return I2C_ERR_INVALID_PARAM;
    }

    // Start write transfer with AUTOEND so STOP is generated by hardware.
    I2C_TransferHandling(I2C_Handle, slave_address, bytesToSend, I2C_AutoEnd_Mode, I2C_Generate_Start_Write);

    // Wait for TXIS flag which is send when the next value can be copied to the data register
    ret = I2C_WaitForFlagOrNackToBeSet(I2C_Handle, I2C_FLAG_TXIS);

    if ((ret & I2C_FLAG_TXIS) != I2C_FLAG_TXIS)
    {
        // got NACK or timeout
        return I2C_ERR_SLAVE_ADDRESS;
    }
```

```c

    /* send data */
    for (idx = 0; idx < bytesToSend; idx++)
    {
        // send data and wait until we get a status back
        I2C_SendData(I2C_Handle, sendData[idx]);

        // Wait on TXIS only until the second to last byte.
        // After writing the last byte, TXIS is not set again.
        if ((idx + 1) < (bytesToSend))
        {
            // wait for TXIS flag which is send when the next value can be copied to the data register
            ret = I2C_WaitForFlagOrNackToBeSet(I2C_Handle, I2C_FLAG_TXIS);
            if ((ret & I2C_FLAG_TXIS) != I2C_FLAG_TXIS)
            {
                // got NACK or timeout
                return I2C_ERR_NACK;
            }
        }
    }

    return I2C_OK;
}

/**
 * @brief  First writes data to a i2c slave device and then reads back the content.
 *         Therefore the first i2c frame is sent without a STOP bit.
 *         The read transfer starts with a repeated START condition.
 * @param  I2C_Handle Pointer to I2C peripheral instance.
 * @param  slave_address Address of the I2C slave device.
 * @param  sendData Data to be sent to the I2C slave device.
 * @param  bytesToSend Number of bytes to send to the I2C slave device.
 * @param  recData Pointer to store received data from the I2C slave device.
 * @param  bytesToRead Number of bytes to read from the I2C slave device.
 * @retval The result of the operation.
 */
I2C_RetType I2C_WriteRead(I2C_TypeDef *I2C_Handle, const uint16_t slave_address, const uint8_t *sendData, const uint8_t
      bytesToSend, uint8_t *recData, const uint8_t bytesToRead)
{
    int32_t idx = 0;
    int32_t ret = 0;

    /* null pointer check */
    if (I2C_Handle == 0 || sendData == 0 || recData == 0 || bytesToRead == 0 || bytesToSend == 0)
    {
        return I2C_ERR_INVALID_PARAM;
    }

    // First frame: write register address without STOP to allow repeated START.
    I2C_TransferHandling(I2C_Handle, slave_address, bytesToSend, I2C_SoftEnd_Mode, I2C_Generate_Start_Write);

    ret = I2C_WaitForFlagOrNackToBeSet(I2C_Handle, I2C_FLAG_TXIS);

    if ((ret & I2C_FLAG_TXIS) != I2C_FLAG_TXIS)
    {
        // got NACK or timeout
        return I2C_ERR_SLAVE_ADDRESS;
    }

    /* send data */
    for (idx = 0; idx < bytesToSend; idx++)
    {
        // send data and wait until we get a status back
        I2C_SendData(I2C_Handle, sendData[idx]);

        // for Last Byte
        if ((idx + 1) == bytesToSend)
        {
            // at last byte Wait for TC Flag
            ret = I2C_WaitForFlagOrNackToBeSet(I2C_Handle, I2C_FLAG_TC);
            if ((ret & I2C_FLAG_TC) != I2C_FLAG_TC)
            {
                // got NACK or timeout
                return I2C_ERR_NACK;
            }
        }
        // only wait on TXIS until the second to last byte
        // Last byte does not get a TXIS!!
        else
        {
            // wait for TXIS flag which is send when the next value can be copied to the data register
            ret = I2C_WaitForFlagOrNackToBeSet(I2C_Handle, I2C_FLAG_TXIS);
            if ((ret & I2C_FLAG_TXIS) != I2C_FLAG_TXIS)
            {
                // got NACK or timeout
                return I2C_ERR_NACK;
            }
        }
    }

    // Second frame: repeated START for read phase, STOP auto-generated at end.
    I2C_TransferHandling(I2C_Handle, slave_address, bytesToRead, I2C_AutoEnd_Mode, I2C_Generate_Start_Read);

    /* receive data */
    for (idx = 0; idx < bytesToRead; idx++)
    {
        // for Reading wait for RXNE Flag
        ret = I2C_WaitForFlagOrNackToBeSet(I2C_Handle, I2C_FLAG_RXNE);

        if ((ret & I2C_FLAG_RXNE) != I2C_FLAG_RXNE)
        {
            // got NACK or timeout
            return I2C_ERR_NACK;
```

```c
        }
        else
        {
            // received data
            recData[idx] = I2C_ReceiveData(I2C_Handle);
        }
    }

    return I2C_OK;
}

/* Private functions ---------------------------------------------------------*/
static int32_t I2C_WaitForFlagOrNackToBeSet(I2C_TypeDef *I2C_Handle, const uint32_t flag)
{
    if(I2C_Handle == 0)
    {
        return I2C_ERR_INVALID_PARAM;
    }

    uint32_t initialTicks = 0;
    uint32_t isrState = 0;

    initialTicks = SysDelay_GetTicks();

    /* Poll until target flag is set, abort early on timeout or NACK. */
    do
    {
        isrState = I2C_ReadRegister(I2C_Handle, I2C_Register_ISR);
        /* check for a timeout */
        if ((SysDelay_GetTicks() - initialTicks) > I2C_TIMEOUT)
        {
            // Return 0 to indicate timeout.
            return 0x00;
        }
        if ((isrState & I2C_FLAG_NACKF) == I2C_FLAG_NACKF)
        {
            return I2C_FLAG_NACKF;
        }
    } while ((isrState & flag) != flag);

    return isrState & flag;
}
```

4.3 board_MCP_temp.h

```c
/**
 ******************************************************************************
 * @file    Board_MCP_Temp.h
 * @author  Simon Offenberger
 * @version V1.0
 * @date    9-March-2026
 * @brief   Driver for the MCP9800 Temperature Sensor on the APUS Board.
 ******************************************************************************
 */

#ifndef __BOARD_MCP_TEMP_H
#define __BOARD_MCP_TEMP_H

/* Includes ------------------------------------------------------------------*/
#include "i2c.h"

/* Defines -------------------------------------------------------------------*/
/** @brief 7-bit MCP9800 slave address shifted for STM32 SPL I2C API. */
#define MCP9800_SLAVE_ADDRESS 0x90

/** @brief MCP9800 temperature register address. */
#define MCP_REG_TEMP 0x00
/** @brief MCP9800 configuration register address. */
#define MCP_REG_CONFIG 0x01
/** @brief MCP9800 hysteresis register address. */
#define MCP_REG_HYSTERESIS 0x02
/** @brief MCP9800 limit register address. */
#define MCP_REG_LIMIT 0x03

#define ONE_SHOT_POS 7
#define ADC_RESOLUTION_POS 5
#define FAULT_QUEUE_POS 3
#define ALERT_POLARITY_POS 2
#define ALERT_MODE_POS 1
#define SHUTDOWN_MODE_POS 0

#define SIGN_BIT_POS 15

#define MCP_MAX_TEMP_CELSIUS 127.5f
#define MCP_MIN_TEMP_CELSIUS -127.5f
/* Exported types ------------------------------------------------------------*/
/**
 * @brief  One-shot conversion mode.
 */
typedef enum
{
  MCP9800_ONE_SHOT_DISABLE = 0,
  MCP9800_ONE_SHOT_ENABLE = 1
} MCP9800_OneShot_Enum;

/**
 * @brief  ADC resolution selection.
 */
typedef enum
{
  MCP9800_ADC_RESOLUTION_0_5C = 0x00,
  MCP9800_ADC_RESOLUTION_0_25C = 0x01,
  MCP9800_ADC_RESOLUTION_0_125C = 0x02,
  MCP9800_ADC_RESOLUTION_0_0625C = 0x03
} MCP9800_Resolution_Enum;

/**
 * @brief  Number of consecutive fault events before alert.
 */
typedef enum
{
  MCP9800_FAULT_QUEUE_1 = 0x00,
  MCP9800_FAULT_QUEUE_2 = 0x01,
  MCP9800_FAULT_QUEUE_4 = 0x02,
  MCP9800_FAULT_QUEUE_6 = 0x03
} MCP9800_FaultQueue_Enum;

/**
 * @brief  Alert output polarity.
 */
typedef enum
{
  MCP9800_ALERT_POLARITY_ACTIVE_LOW = 0,
  MCP9800_ALERT_POLARITY_ACTIVE_HIGH = 1
} MCP9800_AlertPolarity_Enum;

/**
 * @brief  Alert mode selection.
 */
typedef enum
{
  MCP9800_COMPARATOR_MODE = 0,
  MCP9800_INTERRUPT_MODE = 1
} MCP9800_AlertMode_Enum;

/**
 * @brief  Sensor shutdown mode.
 */
typedef enum
{
  MCP9800_SHUTDOWN_MODE_DISABLE = 0,
  MCP9800_SHUTDOWN_MODE_ENABLE = 1
} MCP9800_ShutdownMode_Enum;

/**
```

```c
 * @brief  MCP9800 configuration bitfield values.
 */
typedef struct
{
  MCP9800_ShutdownMode_Enum Shutdown_Mode;
  MCP9800_AlertMode_Enum Alert_Mode;
  MCP9800_AlertPolarity_Enum Alert_Polarity;
  MCP9800_FaultQueue_Enum Fault_Queue;
  MCP9800_Resolution_Enum Resolution;
  MCP9800_OneShot_Enum One_Shot;
} MCP9800_Init_StructTypeDef;

/**
 * @brief  MCP9800 driver handle.
 */
typedef struct
{
  I2C_TypeDef *i2c_handle;
  uint16_t adr;
  MCP9800_Init_StructTypeDef init;

} MCP9800_Temp_HandleTypeDef;

/* Exported functions ------------------------------------------------------- */

/**
 * @brief  Initializes the MCP9800 sensor.
 * @param  handle Pointer to MCP9800 handle structure.
 * @return I2C status.
 */
I2C_RetType MCP9800_Temp_Init(MCP9800_Temp_HandleTypeDef *handle);

/**
 * @brief  Reads the current temperature value.
 * @param  handle Pointer to MCP9800 handle structure.
 * @param  temp Pointer to destination temperature in degree Celsius.
 * @return I2C status.
 */
I2C_RetType MCP9800_Temp_ReadTemp(MCP9800_Temp_HandleTypeDef *handle, float * const temp);

/**
 * @brief  Sets hysteresis temperature threshold.
 * @param  handle Pointer to MCP9800 handle structure.
 * @param  temp Hysteresis threshold in degree Celsius.
 * @return I2C status.
 */
I2C_RetType MCP9800_Temp_Set_THYST(MCP9800_Temp_HandleTypeDef *handle, float temp);

/**
 * @brief  Sets alert temperature threshold.
 * @param  handle Pointer to MCP9800 handle structure.
 * @param  temp Alert threshold in degree Celsius.
 * @return I2C status.
 */
I2C_RetType MCP9800_Temp_Set_TSET(MCP9800_Temp_HandleTypeDef *handle, float temp);

#endif
```

4.4 board_MCP_temp.c

```c
/**
 ******************************************************************************
 * @file    Board_MCP_Temp.c
 * @author  Simon Offenberger
 * @version V1.0
 * @date    9-March-2026
 * @brief   Driver for the MCP9800 Temperature Sensor on the APUS Board.
 ******************************************************************************
 */

/* Includes ------------------------------------------------------------------*/
#include "board_MCP_temp.h"
#include <stdbool.h>

/* Private define ------------------------------------------------------------*/
#define INT_FLOAT_CONV 256.0f
#define ONLY_MSB_BIT 0x80
#define NUM_OF_TEMP_BYTES 2

/**
 * @brief  Initializes the MCP9800 sensor and writes the configuration register.
 * @param  handle Pointer to MCP9800 handle structure.
 * @return I2C status.
 */
I2C_RetType MCP9800_Temp_Init(MCP9800_Temp_HandleTypeDef * handle)
{

  if (handle == 0)
  {
    return I2C_ERR_INVALID_PARAM;
  }

  // Build MCP9800 config register from user-selected bitfields.
  const uint8_t config_reg = ((uint8_t)handle->init.Shutdown_Mode << SHUTDOWN_MODE_POS) |
                             ((uint8_t)handle->init.Alert_Mode << ALERT_MODE_POS) |
                             ((uint8_t)handle->init.Alert_Polarity << ALERT_POLARITY_POS) |
                             ((uint8_t)handle->init.Fault_Queue << FAULT_QUEUE_POS) |
                             ((uint8_t)handle->init.Resolution << ADC_RESOLUTION_POS) |
                             ((uint8_t)handle->init.One_Shot << ONE_SHOT_POS);

  I2C_RetType retval = I2C_Initialize(handle->i2c_handle);

  if (retval != I2C_OK)
  {
    return retval;
  }

  const uint8_t sendData[] = {MCP_REG_CONFIG, config_reg};

  retval = I2C_Write(handle->i2c_handle, handle->adr, sendData, sizeof(sendData) / sizeof(sendData[0]));

  return retval;
}

/**
 * @brief  Reads the current temperature from MCP9800.
 * @param  handle Pointer to MCP9800 handle structure.
 * @param  temp Pointer to destination temperature in degree Celsius.
 * @return I2C status.
 */
I2C_RetType MCP9800_Temp_ReadTemp(MCP9800_Temp_HandleTypeDef *handle, float * const temp)
{
  if (handle == 0 || temp == 0)
  {
    return I2C_ERR_INVALID_PARAM;
  }

  uint8_t recData[NUM_OF_TEMP_BYTES];

  const uint8_t sendData = MCP_REG_TEMP;

  // Write temperature register address, then read back 2 bytes with repeated START.
  I2C_RetType ret = I2C_WriteRead(handle->i2c_handle, handle->adr, &sendData, 1,recData, sizeof(recData) / sizeof(recData[0]))
      ;

  if (ret != I2C_OK)
  {
    return ret;
  }

  // MCP9800 temperature format is Q8.8, convert raw value to degree Celsius.
  // The MSB contains the sign bit and the integer part.
  *temp = (((float)((recData[0] & 0x7F) << 8 | recData[1])) / INT_FLOAT_CONV);

  // Handle sign bit (bit 15) for negative temperatures.
  if((recData[0] & ONLY_MSB_BIT))  {
    *temp *= -1;
  }

  return ret;
}

/**
 * @brief  Sets the MCP9800 overtemperature limit (TSET).
 * @param  handle Pointer to MCP9800 handle structure.
 * @param  temp Temperature limit in degree Celsius.
 * @return I2C status.
 */
I2C_RetType MCP9800_Temp_Set_TSET(MCP9800_Temp_HandleTypeDef *handle, float temp)
{
```

```c

  if (handle == 0 || temp < MCP_MIN_TEMP_CELSIUS || temp > MCP_MAX_TEMP_CELSIUS)
  {
    return I2C_ERR_INVALID_PARAM;
  }

  // Check if temperature is negative
  bool is_negative = temp < 0;

  if(is_negative){
    temp = -temp; // MCP9800 uses a sign bit, so we convert the temperature to positive and set the sign bit later.
  }

  uint16_t temp_int = (uint16_t)(temp * INT_FLOAT_CONV);

  if(is_negative){
    temp_int |= 1<<SIGN_BIT_POS; // Set sign bit for negative temperatures in MCP9800 format.
  }

  // THYST/TSET use MSB plus bit7 of LSB according to MCP9800 register format.
  const uint8_t sendData[] = {MCP_REG_LIMIT, (temp_int >> 8) & 0xFF, temp_int & ONLY_MSB_BIT};

  return I2C_Write(handle->i2c_handle, handle->adr, sendData, sizeof(sendData) / sizeof(sendData[0]));
}

/**
 * @brief  Sets the MCP9800 hysteresis temperature threshold (THYST).
 * @param  handle Pointer to MCP9800 handle structure.
 * @param  temp Hysteresis temperature in degree Celsius.
 * @return I2C status.
 */
I2C_RetType MCP9800_Temp_Set_THYST(MCP9800_Temp_HandleTypeDef *handle, float temp)
{
  if (handle == 0 || temp < MCP_MIN_TEMP_CELSIUS || temp > MCP_MAX_TEMP_CELSIUS)
  {
    return I2C_ERR_INVALID_PARAM;
  }

  // Check if temperature is negative
  bool is_negative = temp < 0;

  if(is_negative){
    temp = -temp; // MCP9800 uses a sign bit, so we convert the temperature to positive and set the sign bit later.
  }

  uint16_t temp_int = (uint16_t)(temp * INT_FLOAT_CONV);

  if(is_negative){
    temp_int |= 1<<SIGN_BIT_POS; // Set sign bit for negative temperatures in MCP9800 format.
  }
  // THYST/TSET use MSB plus bit7 of LSB according to MCP9800 register format.
  const uint8_t sendData[] = {MCP_REG_HYSTERESIS, (temp_int >> 8) & 0xFF, temp_int & ONLY_MSB_BIT};

  return I2C_Write(handle->i2c_handle, handle->adr, sendData, sizeof(sendData) / sizeof(sendData[0]));
}
```